PROJECT DOCUMENTATION

PMFBY District-Level Crop Insurance Performance Analysis

Using Python | 2018–2022

Component	Details
Domain	Agriculture / Crop Insurance
Scheme	Pradhan Mantri Fasal Bima Yojana (PMFBY)
Granularity	District-level, seasonal data
Analysis Period	2018–2022
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

Exploratory Data Analysis, Statistical Analysis and Visualization

1. Project Title

PMFBY District-Level Crop Insurance Performance Analysis Using Python (2018–2022)

The project evaluates farmer participation, seasonal trends, insurance coverage, financial contributions, demographic participation and district performance using PMFBY district-level data.

2. Project Overview

Pradhan Mantri Fasal Bima Yojana (PMFBY) is a crop insurance scheme aimed at providing financial protection to farmers against crop-related risks. This project applies Python-based data analysis to identify meaningful patterns in PMFBY participation and insurance performance.

Technology Stack

Tool	Purpose
Python	Programming and analysis
Pandas	Data cleaning, transformation and aggregation
NumPy	Numerical calculations
Matplotlib	Charts and statistical visualizations
Seaborn	Statistical and categorical plots
Google Colab	Notebook-based development environment

3. Dataset Information

Attribute	Details
Source	India Data Portal / Government of India – PMFBY Data
Location	India (State and District Level)
Timeline	2018–2022
Domain	Agriculture / Crop Insurance
Dataset	Pradhan Mantri Fasal Bima Yojana (PMFBY)
Granularity	District-Level, Seasonal Data
Seasons Covered	Kharif and Rabi
Original Records	6,161
Original Columns	28

4. Problem Definition

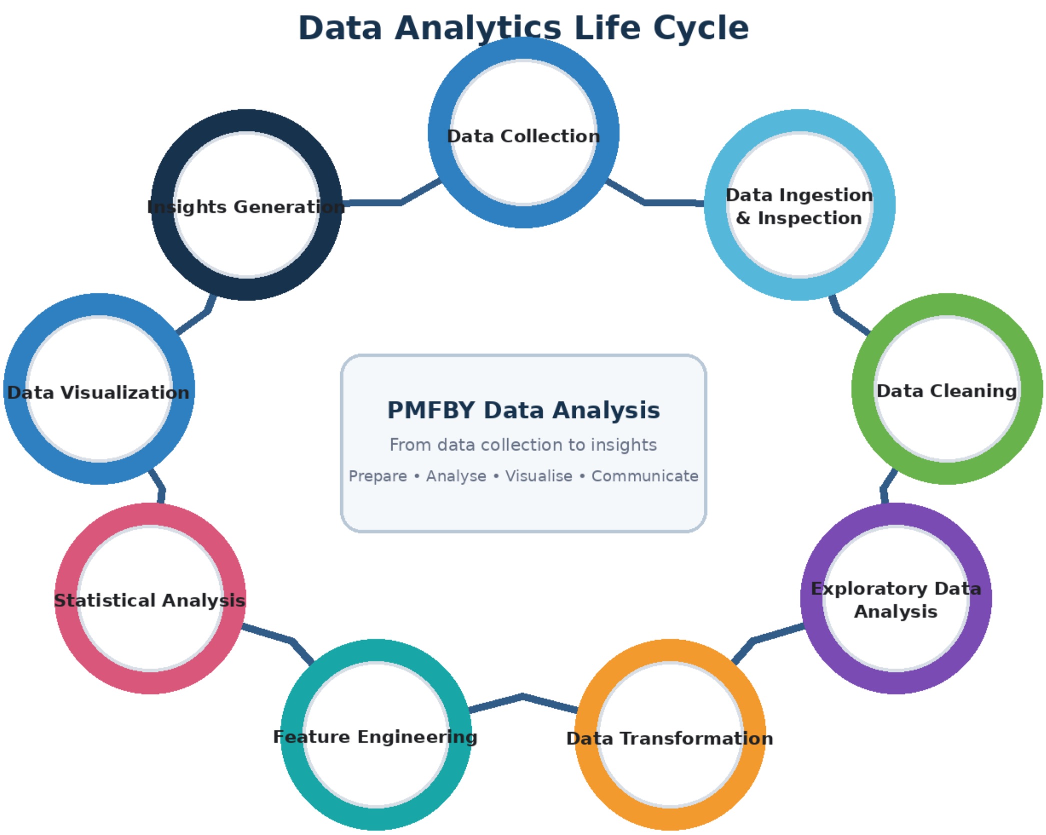
The project analyzes district-level PMFBY data from 2018 to 2022 to evaluate farmer participation, seasonal patterns, crop insurance coverage, premium and financial contributions, and demographic participation. The analysis aims to identify variations across states and districts and derive data-driven insights about participation, insurance coverage and financial contribution patterns.

5. Objectives

- Analyze PMFBY participation across states and districts from 2018–2022.
- Compare PMFBY participation between Kharif and Rabi seasons.
- Analyze the participation pattern of Loanee and Non-Loanee farmers.
- Examine gender and farmer-type participation patterns.
- Evaluate crop insurance coverage using insured area, insurance units and sum insured.
- Analyze gross premium and financial contributions of farmers, State Governments and the Government of India.
- Identify high- and low-performing districts using participation, insurance coverage and financial indicators.

6. Dataset Attributes

Variable Name	Description	Variable Type	Unit of Measurement	Unit Type
ID	Record identifier	Integer	Number	Identifier
Year	Year	Integer	Year	Time
Season	Season	Text	—	Categorical
Scheme	Scheme name	Text	—	Categorical
State_Name	State Name	Text	—	Categorical
District_Name	District Name	Text	—	Categorical
Farmer_Count	Farmers	Numeric	Number	Count
Loanee	Applications by Loanee farmers	Integer	Number	Count
Non_Loanee	Applications by Non-Loanee farmers	Integer	Number	Count
Area_Insured	Area covered	Numeric	Thousand Hectares	Area
Sum_Insured	Sum insured	Numeric	Lakhs	Number
Farmer_Share	Share held by farmers	Numeric	Lakhs	Count
GOI_Share	GOI share	Numeric	Lakhs	Count
State_Share	State share	Numeric	Lakhs	Count
Male	Male farmers	Numeric	Percentage	Percentage
Female	Female farmers	Numeric	Percentage	Percentage
Transgender	Transgender farmers	Numeric	Percentage	Percentage
Marginal	Marginal farmers	Numeric	Percentage	Percentage
Small	Small farmers	Numeric	Percentage	Percentage
Other	Other farmers	Numeric	Percentage	Percentage
Insurance_Unit	Insurance units	Numeric	Number	Count
Gross_Premium	Gross premium	Numeric	Lakhs	Number



7. Data Analytics Life Cycle

Figure: Data Analytics Life Cycle

7.1 Data Ingestion

• The PMFBY dataset was loaded from GitHub into the Google Colab environment using the Pandas library.

• The dataset was stored as a CSV file in the GitHub repository and accessed directly using its raw GitHub URL.

Import Libraries

The required Python library was imported using the following code:

import pandas as pd

Load the Dataset from GitHub

The raw PMFBY dataset was loaded directly from GitHub into a Pandas DataFrame using the read_csv() function.

df = pd.read_csv("GitHub Raw CSV URL")

Dataset Loading

The GitHub-hosted CSV file was directly read into Pandas without manually downloading the dataset to the local system.

The loaded DataFrame was then used for data inspection, data cleaning, preprocessing, feature engineering, data transformation, statistical analysis, and visualization.

7.2 Data Inspection

Inspection	Result / Purpose
Dataset Shape	6,161 rows × 28 columns
First Records	Reviewed using df.head()
Data Types	Reviewed using df.info() and df.dtypes

Statistical Summary	Reviewed using df.describe()
Missing Values	Checked column-wise
Duplicates	Checked; duplicate count was 0

Initial inspection was used to understand the structure, data types, missing values, duplicates and numerical ranges before cleaning.

7.3 Data Cleaning & Preprocessing

- Removed unnecessary columns: state_code, district_code, sc, st, obc and gen.
- Checked ID, Year, Season, Scheme, State and District fields for validity and consistency.
- Standardized District_Name by removing leading and trailing spaces.
- Checked missing Loanee records and imputed the missing value using the relevant district-season median before converting the column to integer.
- Checked negative values for Area_Insured, Sum_Insured, Farmer_Share, GOI_Share, State_Share, Insurance_Unit and Gross_Premium.
- Checked missing demographic and farmer-type values and applied hierarchical median imputation where used in the notebook.
- Checked and handled missing Insurance_Per_Application values after feature creation.

7.4 Feature Engineering

Derived Feature	Formula	Purpose
Total_Applications	Loanee + Non_Loanee	Measure total farmer applications.
Total_Government_Share	GOI_Share + State_Share	Measure total government contribution.
Total_Premium_Contribution	Farmer_Share + GOI_Share + State_Share	Measure total premium contribution.
Insurance_Per_Application	Sum_Insured / Total_Applications	Compare insurance value per application.

7.5 Data Transformation

The cleaned variables were renamed into consistent, analysis-friendly names such as Year, Season, State_Name, District_Name, Loanee, Non_Loanee, Area_Insured, Sum_Insured, Male, Female, Transgender and Insurance_Unit.

7.6 Final EDA Validation

After transformation, the final EDA stage checked remaining null values, zero application records, derived variables and the final dataset structure before statistical analysis and visualization.

8 Statistical Analysis

8.1 Measures of Central Tendency

Variable	Mean	Median	Mode
Total Applications	59,967.75	7,723.00	0.00
Area Insured	41.40	4.30	0.00
Sum Insured	15,993.39	2,554.02	0.00
Gross Premium	2,255.69	202.15	0.00

- **Total Applications:** The mean is 59,967.75, while the median is 7,723.00 and the mode is 0.00. The large difference indicates a right-skewed distribution.
- **Area Insured:** The mean is 41.40, while the median is 4.30 and the mode is 0.00, indicating right skewness.
- **Sum Insured:** The mean is 15,993.39, while the median is 2,554.02 and the mode is 0.00, indicating right skewness.
- **Gross Premium:** The mean is 2,255.69, while the median is 202.15 and the mode is 0.00, indicating right skewness.
- **• Overall:** All four variables show right-skewed distributions, with 0.00 as the mode.

9. Measures of Dispersion – Variance and Standard Deviation

9.1 Variance

Variable	Variance
Total Applications	2.607710×10^{10}
Area Insured	1.560372×10^{4}
Sum Insured	1.088210×10^{9}
Gross Premium	3.828380×10^{7}

Variance indicates considerable dispersion across districts. The variables are measured in squared units, so variance is mainly useful for comparing spread within the same variable.

9.2 Standard Deviation

Variable	Standard Deviation
Total Applications	161,484.05
Area Insured	124.91
Sum Insured	32,988.04
Gross Premium	6,187.39

Standard deviation shows the typical spread in the original unit. Total Applications has a very large spread, while the other variables also show substantial variation across districts.

10. Measures of Distribution – Skewness and Kurtosis

10.1 Skewness

Variable	Skewness
Total Applications	6.442054
Area Insured	11.736275
Sum Insured	4.106434
Gross Premium	5.459718

All four variables have high positive skewness, indicating strongly right-skewed distributions with a small number of observations having exceptionally high values.

10.2 Kurtosis

Variable	Kurtosis
Total Applications	61.225004
Area Insured	249.886077
Sum Insured	27.332569
Gross Premium	38.954620

All four variables have high positive kurtosis, indicating leptokurtic distributions with heavy tails and extreme observations.

11. Univariate Analysis

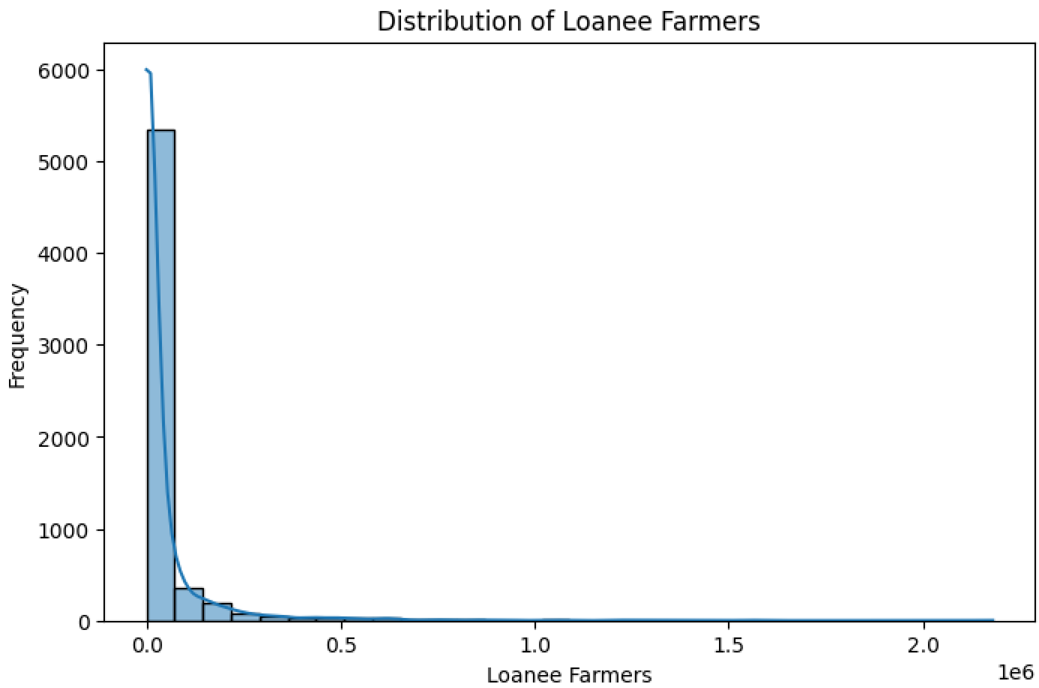
11.1 Loanee Farmer Distribution

Objective: To analyse the distribution of Loanee farmer participation.

Attribute: Loanee

Chart Type: Histogram with Kernel Density Estimation (KDE)

Plot Summary: Analyzes the frequency and density distribution of Loanee farmer participation across the dataset, highlighting the concentration of observations at lower participation levels and the long right tail.



- • **Key Insight:** Most observations are concentrated in the lower Loanee participation range.
- • **Pattern:** The distribution has a long right tail, showing that a small number of observations have very high Loanee participation.
- • **Observation:** Loanee participation shows considerable variation across observations.
- • **Distribution:** The histogram is positively/right-skewed.
- • **Overall:** Loanee farmer participation is unevenly distributed, with a small number of observations recording exceptionally high participation.

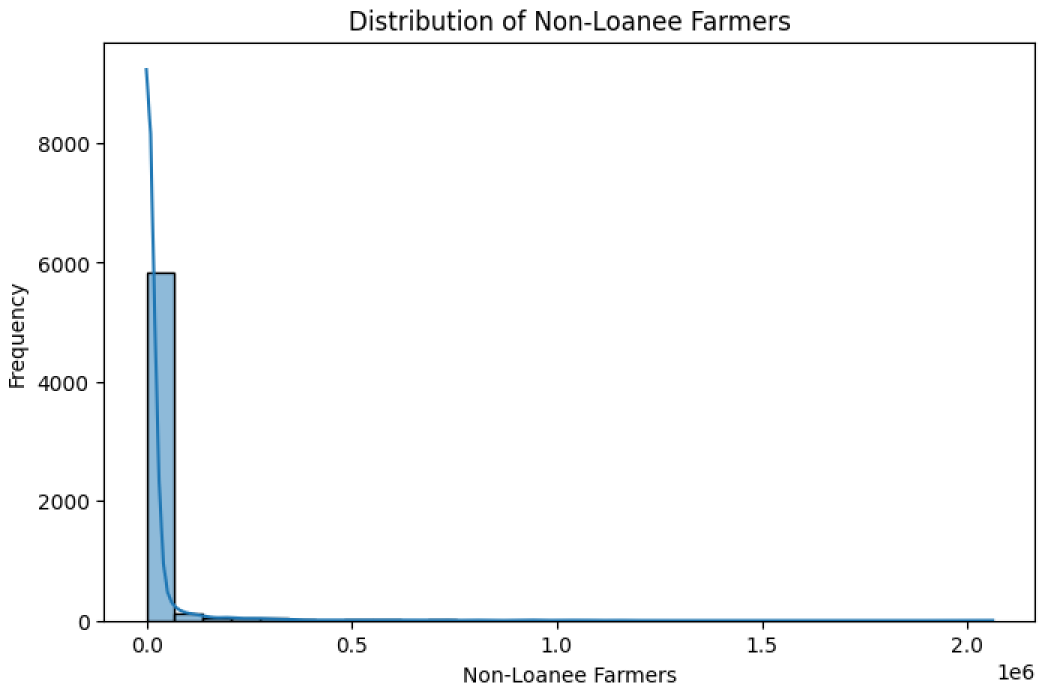
11.2 Non-Loanee Farmer Distribution

Objective: To analyse the distribution of Non-Loanee farmer participation.

Attribute: Non_Loanee

Chart Type: Histogram with Kernel Density Estimation (KDE)

Plot Summary: Analyzes the frequency and density distribution of Non-Loanee farmer participation across the dataset, highlighting the concentration of observations at lower participation levels and the long right tail.



- • **Key Insight:** Most observations are concentrated in the lower Non-Loanee participation range.
- • **Pattern:** The distribution has a long right tail, showing that a small number of observations have very high Non-Loanee participation.
- • **Observation:** Non-Loanee participation shows considerable variation across observations.
- • **Distribution:** The histogram is positively/right-skewed.
- • **Overall:** Non-Loanee farmer participation is unevenly distributed, with a small number of observations recording exceptionally high participation.

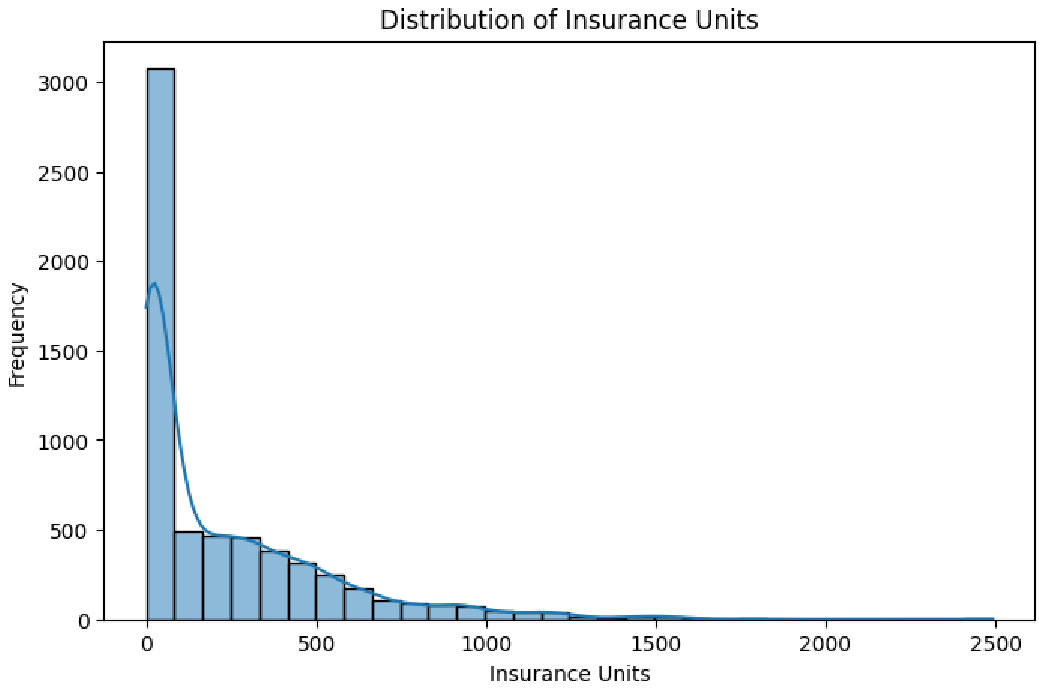
11.3 Insurance Unit Distribution

Objective: To understand the distribution of insurance units across the observations.

Attribute: Insurance_Unit

Chart Type: Histogram with Kernel Density Estimation (KDE)

Plot Summary: Analyzes the frequency and density distribution of Insurance Unit values across the dataset, highlighting the concentration of observations at lower values and the positively skewed distribution.



- • **Key Insight:** Most observations are concentrated in the lower Insurance Unit range.
- • **Pattern:** The distribution has a long right tail, showing that a small number of observations have substantially higher Insurance Unit values.
- • **Observation:** Insurance Unit values show considerable variation across observations.
- • **Distribution:** The histogram is positively/right-skewed.
- • **Overall:** Insurance Unit coverage is unevenly distributed, with a small number of observations recording exceptionally high values.

12. Bivariate Analysis

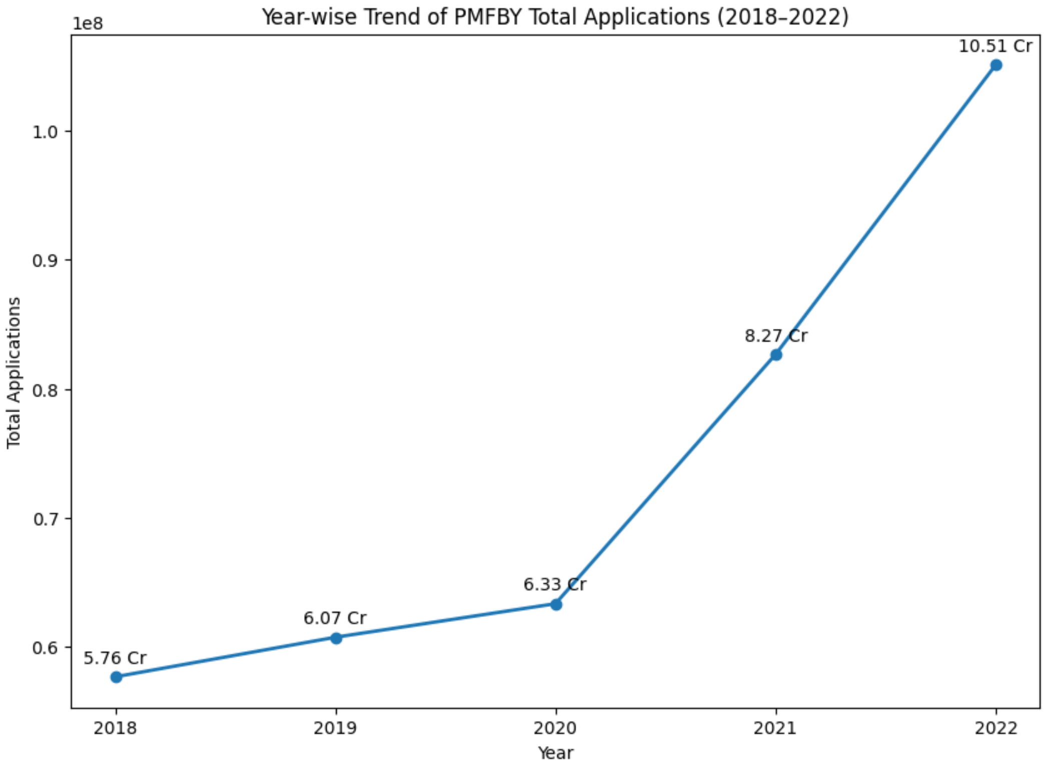
12.1 Year-wise Trend of Total Applications

Objective: To analyse the yearly trend in PMFBY participation.

Attribute: Year and Total_Applications

Chart Type: Line Chart

Plot Summary: Analyzes the year-wise trend of Total Applications and shows the overall change in PMFBY participation across the study years.



- • **Key Insight:** Total Applications show a continuous increasing trend from 2018 to 2022.
- • **Pattern:** Applications increased gradually from 2018 to 2020, followed by a sharper increase in 2021 and 2022.
- • **Observation:** The highest Total Applications were recorded in 2022, while the lowest were recorded in 2018.
- • **Trend:** The chart indicates a strong upward trend in PMFBY participation.
- • **Overall:** PMFBY participation increased substantially between 2018 and 2022, with the most significant growth occurring during 2020–2022.

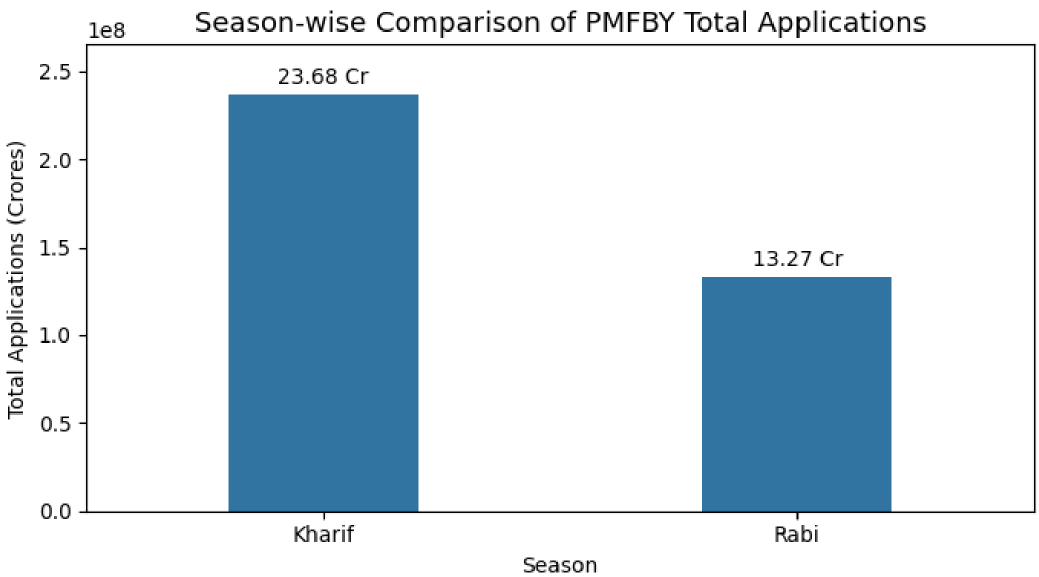
12.2 Season-wise Total Applications

Objective: To compare PMFBY participation across different seasons.

Attribute: Season and Total_Applications

Chart Type: Bar Chart

Plot Summary: Compares Total Applications across Kharif and Rabi seasons and highlights the difference in seasonal participation.



- • **Key Insight:** Kharif recorded approximately 23.5 Cr Total Applications, while Rabi recorded approximately 13.3 Cr.
- • **Pattern:** Kharif participation is considerably higher than Rabi.
- • **Observation:** Kharif participation is approximately 1.8 times higher than Rabi participation.
- **Comparison: The difference indicates a strong seasonal variation in PMFBY participation.**
- • **Overall:** Kharif is the more dominant season for PMFBY participation.

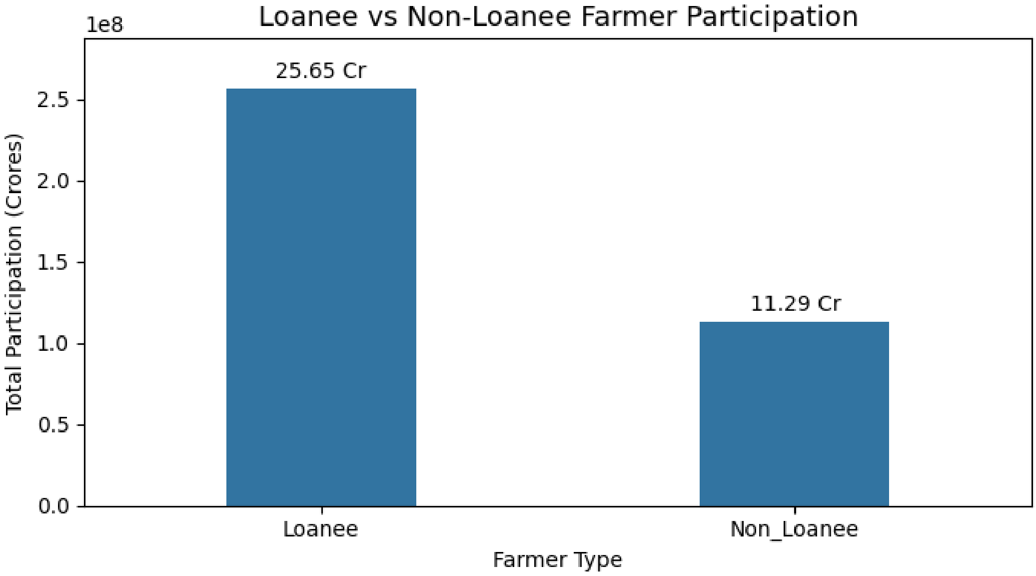
12.3 Loanee vs Non-Loanee Participation

Objective: To compare Loanee and Non-Loanee farmer participation.

Attribute: Loanee and Non_Loanee

Chart Type: Bar Chart

Plot Summary: Compares Loanee and Non-Loanee participation and highlights the difference between the two farmer groups.



- • **Key Insight:** Loanee farmers recorded approximately 25.7 Cr participation, while Non-Loanee farmers recorded approximately 11.3 Cr.
- • **Pattern:** Loanee participation is considerably higher than Non-Loanee participation.
- • **Observation:** Loanee participation is approximately 2.3 times higher than Non-Loanee participation.
- **Comparison: The chart shows a clear participation gap between the two groups.**
- • **Overall:** Loanee farmers have a significantly higher participation level than Non-Loanee farmers under PMFBY.

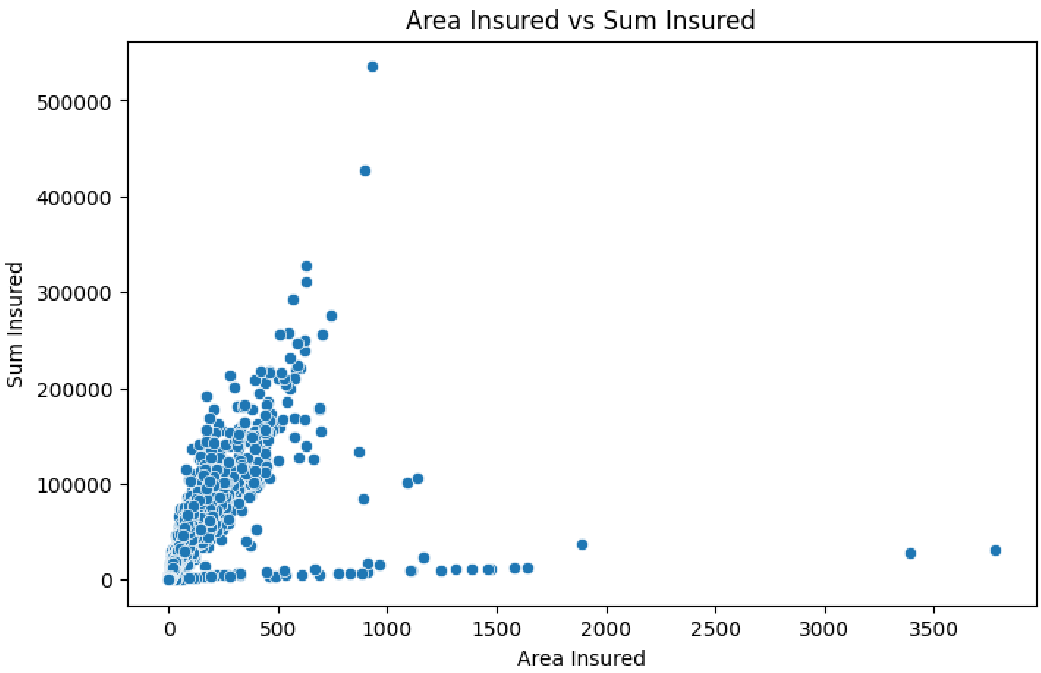
12.4 Area Insured vs Sum Insured

Objective: To analyse the relationship between Area Insured and Sum Insured.

Attribute: Area_Insured and Sum_Insured

Chart Type: Scatter Plot

Plot Summary: Analyzes the relationship between Area Insured and Sum Insured and shows how insurance value varies with insured area.



- • **Key Insight:** The scatter plot shows a positive relationship between Area Insured and Sum Insured.
- • **Pattern:** Most observations are concentrated below approximately 800 Area Insured and 2.5 lakh Sum Insured.
- • **Observation:** A few observations have exceptionally high Sum Insured values, including values above 3 lakh, with the highest reaching approximately 5.3 lakh.
- **Variation: Some observations have relatively high Area Insured but low Sum Insured, so the relationship is not perfectly consistent.**
- • **Overall:** Larger insured areas generally correspond to higher Sum Insured, with considerable variation across observations.

13. Multivariate Analysis

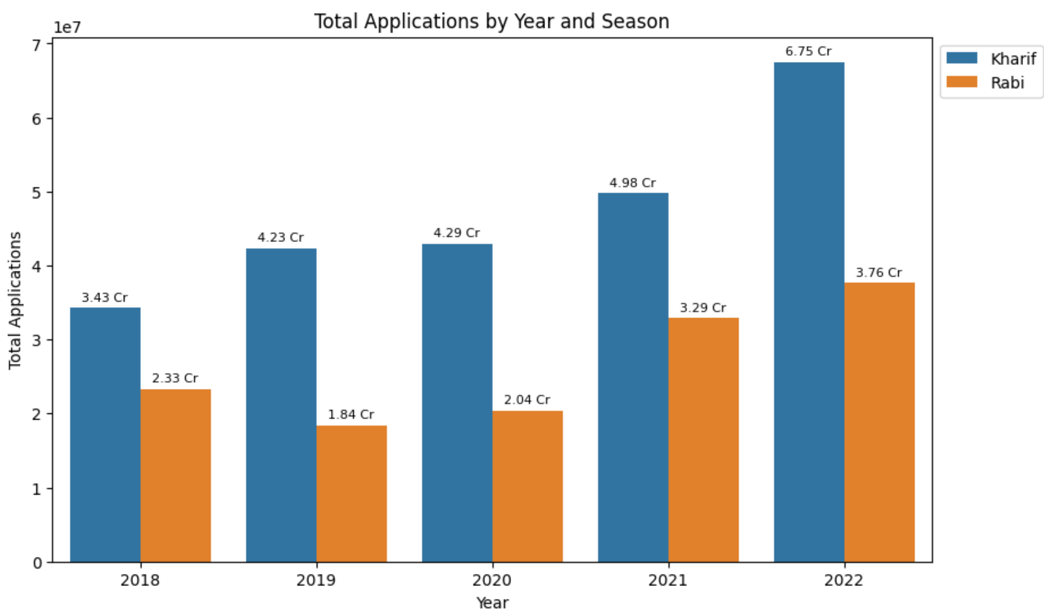
13.1 Year-wise and Season-wise Total Applications

Objective: To analyse PMFBY participation by year and season.

Attribute: Year, Season and Total_Applications

Chart Type: Grouped Bar Chart

Plot Summary: Compares Kharif and Rabi Total Applications across years to identify yearly and seasonal participation patterns.



- • **Key Insight:** Kharif applications increased from 3.43 Cr in 2018 to 6.75 Cr in 2022, while Rabi applications increased from 2.33 Cr to 3.76 Cr.
- • **Pattern:** Kharif participation remained higher than Rabi in every year from 2018 to 2022.
- • **Trend:** Both Kharif and Rabi show an overall increasing trend, with the most noticeable growth occurring between 2020 and 2022.
- • **Observation:** The highest Kharif applications (6.75 Cr) and highest Rabi applications (3.76 Cr) were recorded in 2022.
- • **Overall:** PMFBY participation increased over the years in both seasons, but Kharif remained the dominant season.

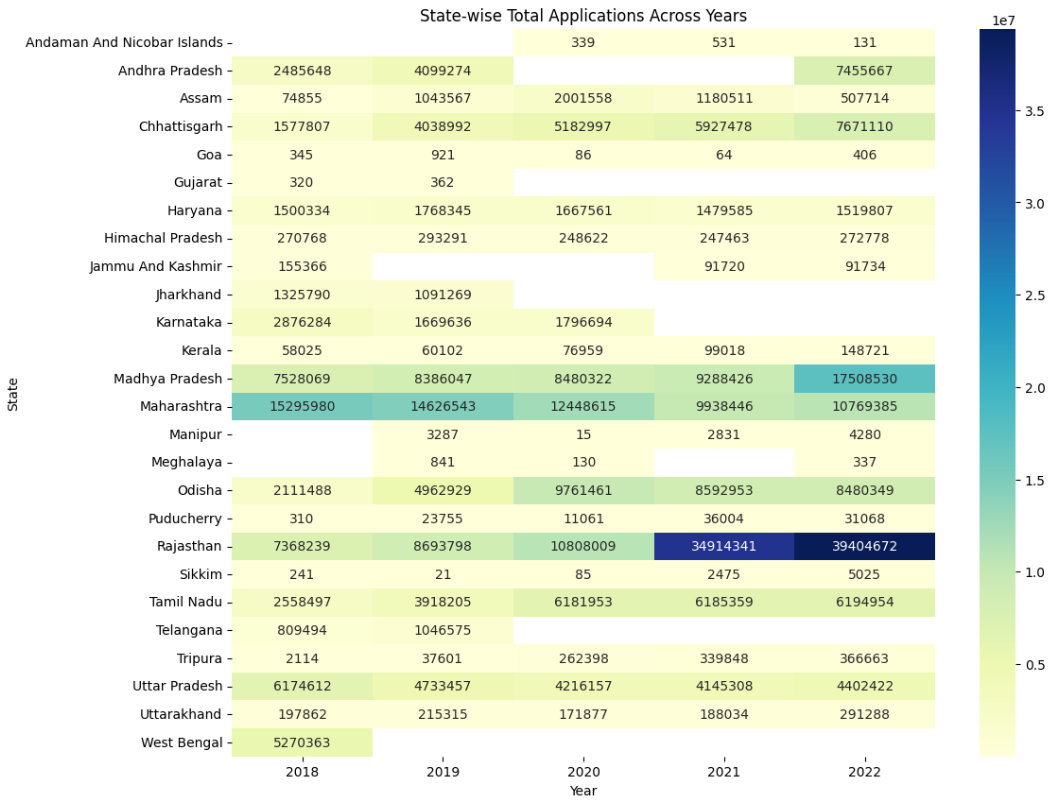
13.2 State-wise Total Applications Across Years

Objective: To compare state-wise PMFBY participation across years.

Attribute: State_Name, Year and Total_Applications

Chart Type: Heatmap

Plot Summary: Analyzes state-wise Total Applications across years and highlights differences in participation using color intensity.



- **Color Intensity: Darker shades indicate higher Total Applications, while lighter shades indicate lower participation.**
- • **Key Insight:** Rajasthan recorded the highest visible participation, increasing from approximately 7.37 million in 2018 to 39.40 million in 2022.
- • **Pattern:** Rajasthan shows particularly strong growth during 2021 and 2022.
- • **Observation:** Madhya Pradesh also shows strong participation, increasing from approximately 7.53 million in 2018 to 17.51 million in 2022.
- • **Trend:** Tamil Nadu shows an overall increase, from approximately 2.56 million in 2018 to 6.19 million in 2022.
- • **Overall:** The heatmap shows substantial variation across states and years.

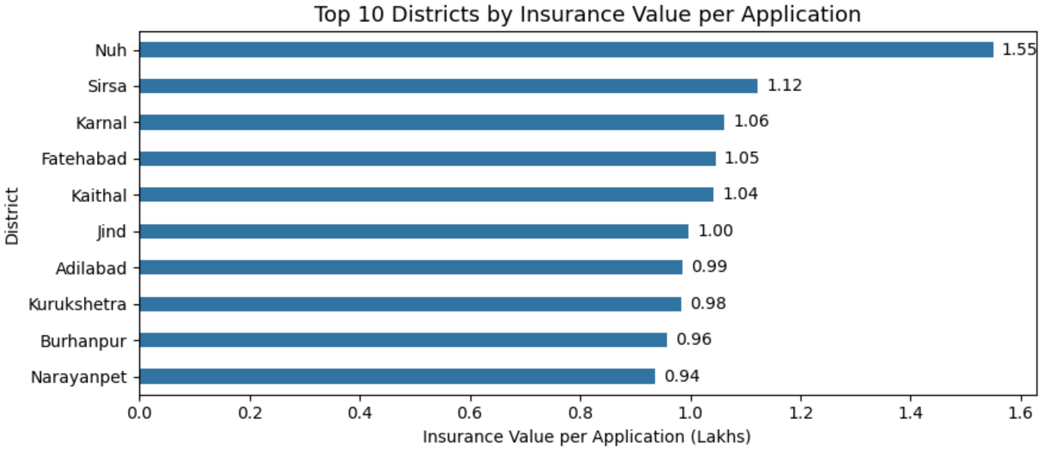
13.3 Top 10 Districts by Insurance Value per Application

Objective: To identify top-performing districts based on insurance value per application.

Attribute: District_Name and Insurance_Per_Application

Chart Type: Horizontal Bar Chart

Plot Summary: Ranks the top 10 districts by Insurance Value per Application and highlights differences in insurance value across districts.



- • **Key Insight:** Nuh has the highest Insurance Value per Application at approximately 1.55 Lakhs, followed by Sirsa (1.12 Lakhs) and Karnal (1.06 Lakhs).
- • **Pattern:** The value decreases from 1.55 Lakhs in Nuh to 0.94 Lakhs in Narayanpet.
- • **Observation:** Sirsa has 10.87 Lakhs total applications, while Nuh has 0.30 Lakhs applications but the highest Insurance Value per Application.
- **Comparison: A district with a high number of applications does not necessarily have the highest Insurance Value per Application.**
- • **Overall:** Nuh is the highest-performing district based on Insurance Value per Application among the selected top 10.

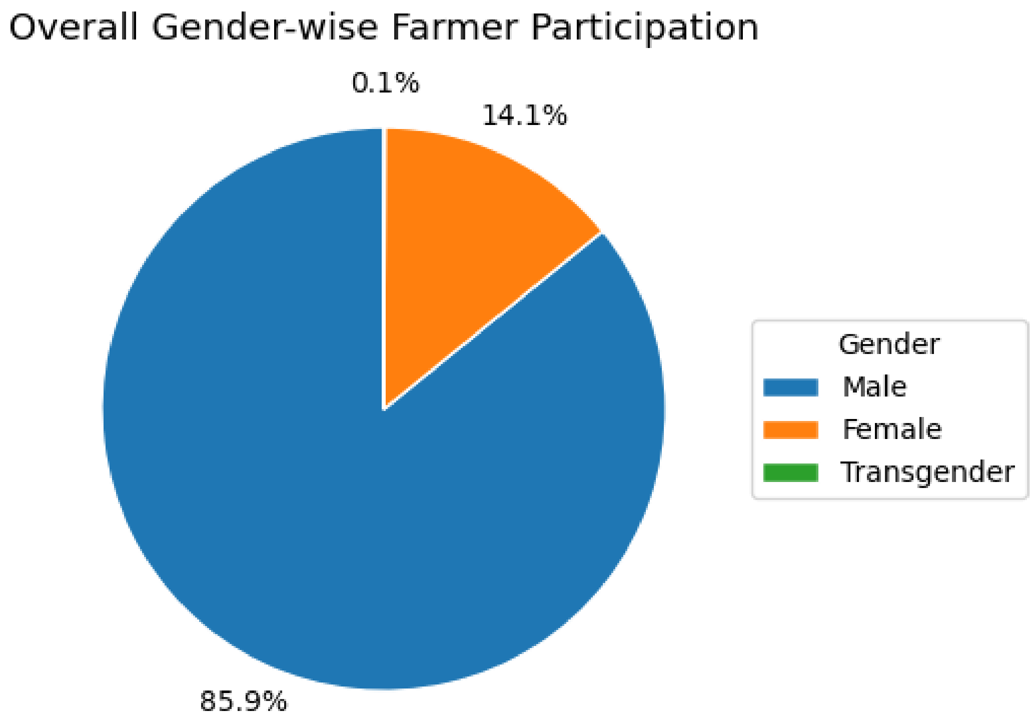
13.4 Overall Gender-wise Farmer Participation

Objective: To understand the overall gender-wise participation share.

Attribute: Male, Female and Transgender

Chart Type: Pie Chart

Plot Summary: Shows the proportion of farmer participation by gender and highlights the relative participation share of each gender group.



- • **Key Insight:** Male farmers account for 85.9% of participation, Female farmers 14.1%, and Transgender farmers 0.1%.
- • **Pattern:** Male participation is significantly higher than Female and Transgender participation.
- • **Observation:** Female participation represents a smaller but noticeable share of overall participation.
- **Comparison: There is a large participation gap between Male and Female farmers, while Transgender participation is very small.**
- • **Overall:** PMFBY farmer participation is predominantly male.

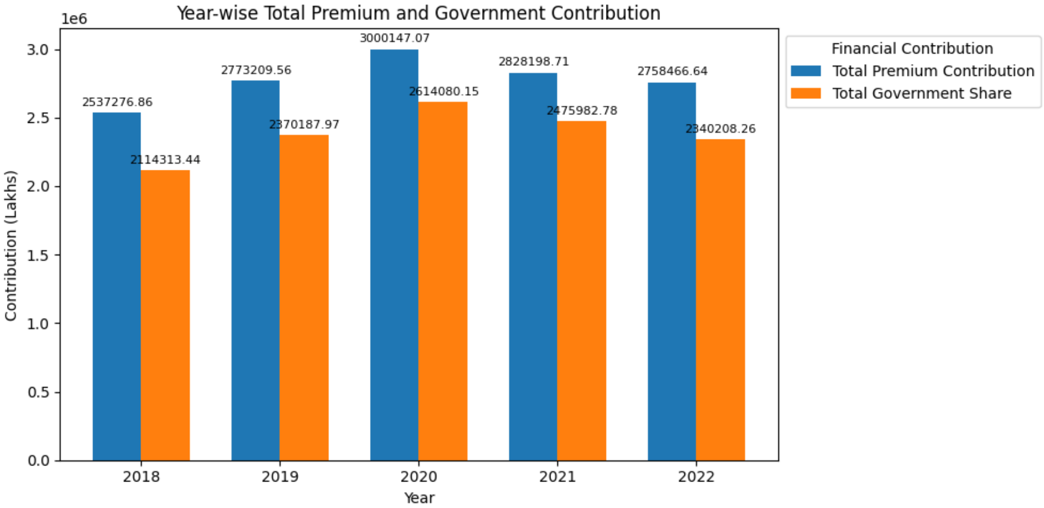
13.5 Year-wise Total Premium and Government Contribution

Objective: To analyse yearly premium contribution and government financial support.

Attribute: Year, Total_Premium_Contribution and Total_Government_Share

Chart Type: Line Chart

Plot Summary: Analyzes yearly Total Premium Contribution and Government Share to understand changes in premium contribution and government support.



- • **Key Insight:** Total Premium Contribution is higher than Total Government Share in every year.
- • **Pattern:** Total Premium Contribution increased from 25.37 Lakhs in 2018 to 30.00 Lakhs in 2020.
- • **Observation:** Total Government Share increased from 21.14 Lakhs in 2018 to 26.14 Lakhs in 2020.
- • **Trend:** After reaching a peak in 2020, both measures declined in 2021 and 2022.
- **Comparison: In 2022, Total Premium Contribution was approximately 27.58 Lakhs, while Total Government Share was approximately 23.40 Lakhs.**
- • **Overall:** Both financial measures increased until 2020 and then declined.

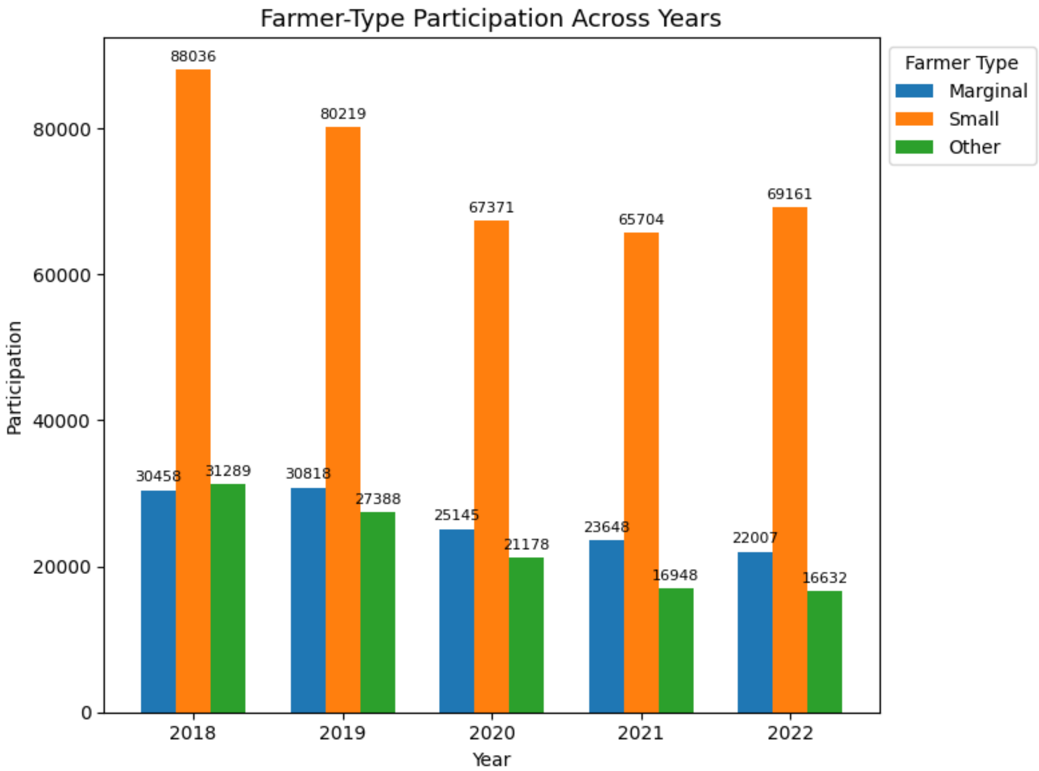
13.6 Farmer-Type Participation Across Years

Objective: To analyse farmer-type participation across years.

Attribute: Year, Small, Marginal and Other

Chart Type: Line Chart

Plot Summary: Compares Small, Marginal and Other farmer participation across years to identify differences and trends among farmer groups.



- • **Key Insight:** Small farmers have the highest participation in every year, followed by Marginal farmers, while Other farmers have the lowest participation.
- • **Pattern:** Small farmer participation decreased from 88,036 in 2018 to 65,704 in 2021, followed by an increase to 69,161 in 2022.
- • **Observation:** Marginal farmer participation declined from 30,458 in 2018 to 22,007 in 2022.
- • **Trend:** Other farmer participation declined from 31,289 in 2018 to 16,632 in 2022.
- • **Overall:** Small farmers are the dominant participating group, while Marginal and Other participation generally declined.

14. Key Findings

- **Overall PMFBY Participation Growth:** Total Applications show a strong overall upward trend from 2018 to 2022, with the highest participation recorded in 2022.
- **Seasonal Participation:** Kharif remained higher than Rabi in every analyzed year. Kharif applications increased from 3.43 Cr in 2018 to 6.75 Cr in 2022, while Rabi increased from 2.33 Cr to 3.76 Cr.
- **State and District Variation:** PMFBY participation varies substantially across states and districts, showing clear regional differences in scheme participation over the study period.
- Loanee and Non-Loanee Participation: Loanee participation is substantially higher than Non-Loanee participation, indicating a clear difference between the two farmer groups.
- **Gender Participation:** Male farmers account for 85.9% of overall participation, while Female farmers account for 14.1% and Transgender farmers account for 0.1%.
- **Farmer-Type Participation:** Small farmers recorded the highest participation across the analyzed years, followed by Marginal and Other farmers.
- **Insurance Coverage:** Insurance-related measures such as Area Insured, Insurance Unit and Sum Insured show considerable variation across observations. Area Insured and Sum Insured also show a positive relationship.
- **Financial Contributions:** Total Premium Contribution remained higher than Total Government Share across the analyzed years. Both financial measures reached their highest levels around 2020 and declined afterward.
- **District Performance:** Nuh recorded the highest Insurance Value per Application at 1.55 Lakhs among the selected top 10 districts, followed by Sirsa at 1.12 Lakhs and Karnal at 1.06 Lakhs.

- **Distribution and Variability:** Major numerical variables show strong positive skewness and high variability, indicating many lower-value observations alongside a smaller number of high-value observations.

15. Analytical Framework

15.1 Descriptive Analysis — What Happened?

The analysis summarizes PMFBY participation, seasonal trends, insurance coverage, farmer demographics, district performance and financial contributions using descriptive statistics and visualizations.

15.2 Diagnostic Analysis — Why Did It Happen?

The analysis shows clear differences across seasons, regions and farmer groups. Loanee, gender and farmer-type participation vary substantially, while district-level Insurance Value per Application also differs considerably. High skewness and kurtosis indicate strong variation and extreme observations. These are observed patterns, not confirmed causal relationships.

15.3 Predictive Analysis — What May Happen?

Historical trends can be used to estimate future PMFBY applications, seasonal participation, state and district participation, insurance coverage and financial contributions. Actual prediction requires a suitable predictive model and validation.

15.4 Prescriptive Analysis — What Should Be Done?

The findings suggest monitoring low-participation regions and farmer groups, strengthening Non-Loanee and Female farmer outreach, studying high-performing districts such as Nuh, Sirsa and Karnal, and monitoring seasonal, farmer-type and financial trends. Data quality and extreme values should be validated before advanced modelling.

16. Recommendations and Decision Support

Improve Data Quality Controls

Validate missing values, duplicates, inconsistent names, zero values and extreme observations before advanced analysis.

Focus on Low-Participation Regions

Investigate districts and states with comparatively lower participation to identify possible coverage or outreach gaps.

Strengthen Non-Loanee and Female Participation

Use targeted awareness and accessibility initiatives to address the participation gaps identified in the analysis.

Monitor Seasonal and Farmer-Type Trends

Track Kharif/Rabi patterns and changes among Small, Marginal and Other farmers.

Investigate District Performance

Study high Insurance Value per Application districts to understand factors associated with stronger insurance coverage.

Monitor Financial Contributions

Track Farmer, Government and Total Premium Contribution trends over time.

17. Conclusion and Future Scope

The PMFBY analysis provides a structured understanding of farmer participation, seasonal behaviour, insurance coverage, demographic participation, district performance and financial contribution patterns from 2018 to 2022.

Future Scope

- Automate the Python analysis workflow for future PMFBY dataset updates.
- Generate automated analytical reports after data refresh.
- Apply predictive models to forecast future participation.
- Extend the study with crop, weather and additional agricultural variables.